LAB WORKSHEET

ERD Designer

Design a database model in a Visio-style tool: entities, keys, relationships, constraints and SQL.

Key vocabulary

- **PK** uniquely identifies a row · **FK** must match the parent's PK — draw relationships from **parent → child**.
- **NOT NULL** = required · **UNIQUE** = no duplicates (e.g. email) · **DEFAULT** e.g. `'Booked'`, `CURRENT_DATE` · **CHECK** e.g. `Price >= 0`, `Status IN ('Booked','Done')`.
- Money is `DECIMAL(6,2)`; phone numbers are `VARCHAR`.
- Cardinality `1`, `0..1`, `0..*`, `1..*`; the FK sits on the many side.
- ON DELETE: **CASCADE** deletes children · **RESTRICT** blocks the delete · **SET NULL** clears the FK (it must allow NULL).
- Many-to-many → a link entity with a composite PK made of both FKs (identifying relationships).

How the lab works

1. **Guided build**: build each step, press **Check this step**, fix it from the feedback. Cycle Hire → Sixth Form clubs (link entities) → Vet Surgery (optional relationships, SET NULL).
2. **Independent task**: Pennine Pets Grooming — press **Check my model** until every requirement is ✓.
3. **Free mode**: five client briefs marked requirement by requirement, plus a blank canvas and a bug-hunt example.

Always finish with **Validate model** (zero errors), **Generate SQL**, and prove your constraints in the **Constraint tester**.